

JAVA LAB RECORD

Experiment No: 04

Date: 20/09/26

Title: Student Performance Analysis System Using Java

1. Objective / Aim

To design and implement a Java-based Student Performance Analysis System using object-oriented programming, arrays, constructors, methods, loops, conditional statements, and String operations to calculate total, average, highest and lowest marks, percentage, grade, result, and performance remarks.

2. System Requirements

Hardware: Computer/Laptop, Intel Core i3 or equivalent processor, minimum 4 GB RAM, minimum 1 GB free storage, keyboard and monitor.

Software: Windows/Linux/macOS, JDK 17 or 21, Eclipse/IntelliJ IDEA/NetBeans/VS Code, Command Prompt/Terminal, javac and java.

3. Algorithm / Logic Description

1. Start the program.
2. Create a Scanner object to accept input from the user.
3. Read the student's roll number and name.
4. Create an integer array of size 5 to store marks for five subjects.
5. Read the five subject marks using a for loop.
6. Create a Student object by passing roll number, name, and marks to the constructor.
7. Calculate total marks by adding all elements of the marks array.
8. Calculate average by dividing total marks by the number of subjects.
9. Find the highest and lowest marks.
10. Calculate percentage using total marks and the maximum possible marks.
11. Determine grade using if-else conditions.
12. Determine PASS or FAIL using a 50% threshold.
13. Generate a performance remark based on the percentage.
14. Trim and convert the student name to uppercase and display its length.
15. Display the complete performance report and close the Scanner.
16. Stop the program.

4. UML Class Diagram

```
+-----+
| Student |
+-----+
| - rollNumber : int |
| - studentName : String |
| - marks : int[] |
+-----+
| + Student(int, String, int[]) |
+-----+
```

```
| + calculateTotal() : int |
| + calculateAverage() : double |
| + findHighest() : int |
| + findLowest() : int |
| + calculatePercentage() : double |
| + determineGrade() : String |
| + determineResult() : String |
| + performanceRemark() : String |
| + displayDetails() : void |
+-----+
                creates
StudentPerformanceAnalysis -----> Student
+-----+
| + main(String[]) : void |
+-----+
```

5. Source Code

```
import java.util.Scanner;

class Student {
    int rollNumber;
    String studentName;
    int[] marks;

    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    int calculateTotal() {
        int total = 0;
        for (int mark : marks) {
            total += mark;
        }
        return total;
    }

    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    int findHighest() {
        int highest = marks[0];
        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }
        return highest;
    }

    int findLowest() {
        int lowest = marks[0];
        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }
        return lowest;
    }

    double calculatePercentage() {
        int total = calculateTotal();
        return (total / (marks.length * 100.0)) * 100;
    }

    String determineGrade() {
        double percentage = calculatePercentage();
        if (percentage >= 90)
            return "A+";
        else if (percentage >= 80)
            return "A";
        else if (percentage >= 70)
            return "B";
        else if (percentage >= 60)
            return "C";
        else if (percentage >= 50)
            return "D";
        else
            return "F";
    }

    String determineResult() {
        if (calculatePercentage() >= 50)
            return "PASS";
        else
            return "FAIL";
    }

    String performanceRemark() {
        double percentage = calculatePercentage();
    }
```

```

        if (percentage >= 90)
            return "Excellent Performance";
        else if (percentage >= 80)
            return "Very Good Performance";
        else if (percentage >= 70)
            return "Good Performance";
        else if (percentage >= 60)
            return "Average Performance";
        else if (percentage >= 50)
            return "Needs Improvement";
        else
            return "Poor Performance";
    }

    void displayDetails() {
        String formattedName = studentName.trim().toUpperCase();
        double percentage = calculatePercentage();
        double roundedPercentage =
            Math.round((percentage * 100.0) / 100.0);

        System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");
        System.out.println("Roll Number      : " + rollNumber);
        System.out.println("Student Name   : " + formattedName);
        System.out.println("Name Length    : " + formattedName.length());
        System.out.println("Marks:");
        for (int i = 0; i < marks.length; i++) {
            System.out.println("Subject " + (i + 1) + "      : " + marks[i]);
        }
        System.out.println("Total Marks      : " + calculateTotal());
        System.out.println("Average Marks    : " + calculateAverage());
        System.out.println("Highest Marks    : " + findHighest());
        System.out.println("Lowest Marks     : " + findLowest());
        System.out.println("Percentage       : " + roundedPercentage + "%");
        System.out.println("Grade            : " + determineGrade());
        System.out.println("Result           : " + determineResult());
        System.out.println("Performance      : " + performanceRemark());
        System.out.println("=====");
    }
}

public class StudentPerformanceAnalysis {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        int[] marks = new int[5];
        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < marks.length; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        Student student =
            new Student(rollNumber, studentName, marks);

        student.displayDetails();
        sc.close();
    }
}

```

6. Completion and Execution Commands

1. Save the program as:
StudentPerformanceAnalysis.java
2. Compile:
javac StudentPerformanceAnalysis.java
3. Execute:
java StudentPerformanceAnalysis

7. Modified Output

```
Enter Student Name: trivikram
Enter marks for 5 subjects:
Subject 1: 90
Subject 2: 80
Subject 3: 95
Subject 4: 92
Subject 5: 89

===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 251
Student Name     : TRIVIKRAM
Name Length      : 9
Marks:
Subject 1        : 90
Subject 2        : 80
Subject 3        : 95
Subject 4        : 92
Subject 5        : 89
Total Marks      : 446
Average Marks    : 89.2
Highest Marks    : 95
Lowest Marks     : 80
Percentage       : 89.2%
Grade            : A
Result           : PASS
Performance      : Very Good Performance
=====

C:\Users\TRIVIKRAM>SS
```

Result: The Student Performance Analysis System was successfully designed, compiled, and executed with the modified student output shown above.